

AI FOR SUSTAINABILITY

Pasumai Madurai (பசுமை மதுரை)

An AI-Powered Waste Segregation Assistant for a Sustainable City

DEVELOPED UNDER

1M1B Virtual Internship • IBM SkillsBuild • AICTE Initiative

Meenakshi Temple Madurai vector gopuram

■ Introduction & Mission

Empowering Citizens, Protecting Workers

Our primary mission is to foster clean, efficient, and proactive waste sorting at the household level across Madurai. By embedding smart tech locally, we transform citizens from passive dumpers to active conservationists.

CORE OBJECTIVE

Dignify and ease the immense daily sorting burden on Madurai's dedicated Thoimai Paniyalar (sanitation workers).



SDG GOAL 11

Sustainable Cities & Communities

Creating safer, cleaner municipal habitats.



SDG GOAL 12

Responsible Consumption & Production

Optimizing systemic reuse and circular streams.

■ The Challenge of Household Waste

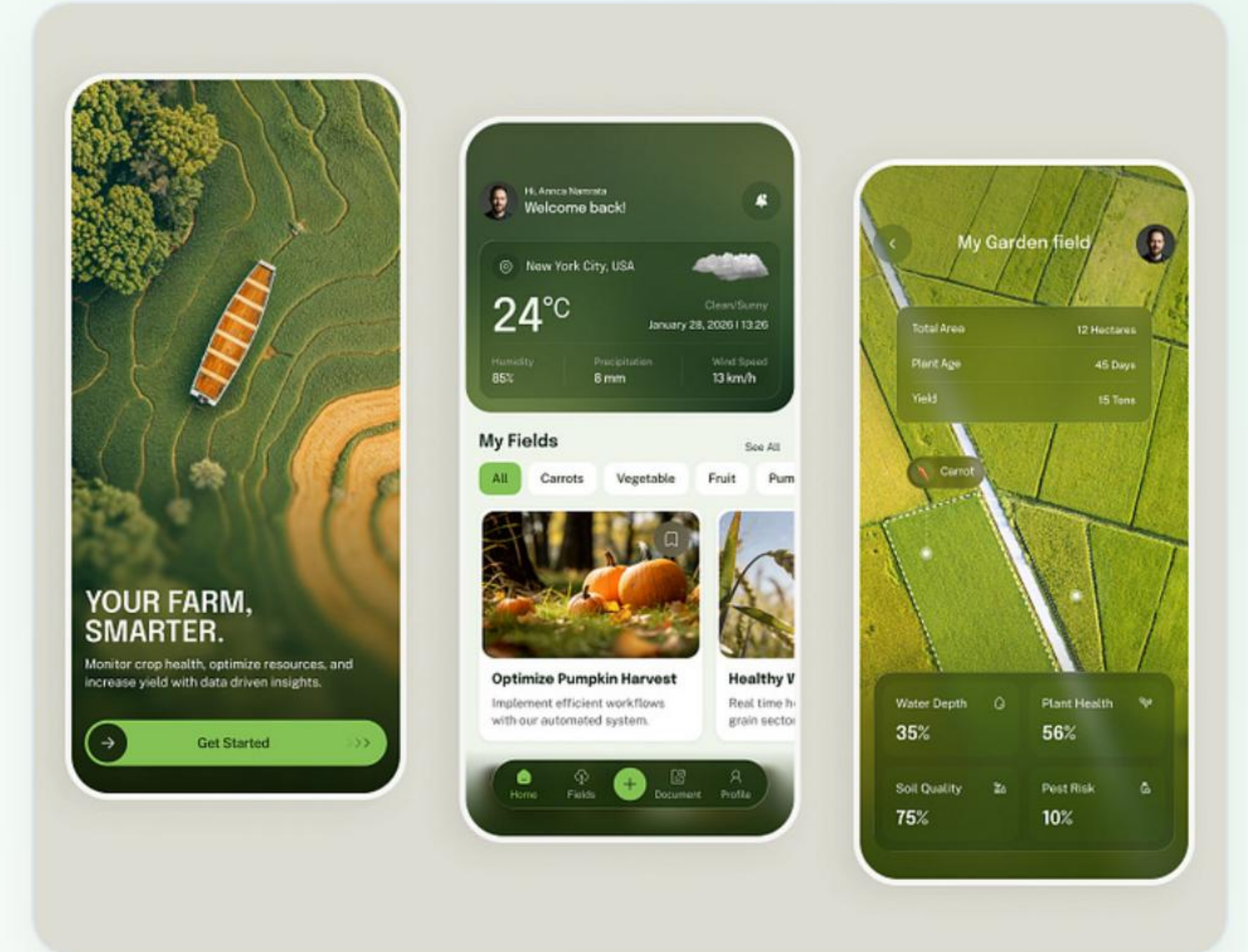
-  **Disposal Uncertainty:** Citizens lack easily accessible, clean awareness routes for sorting complex domestic waste.
-  **Mixed Stream Pollution:** Combined garbage leads directly to toxic ground leaching and severely choked urban drainage.
-  **Methane Generation:** Trapped food and organic waste in massive unsegregated landfills accelerate high gas emissions.
-  **Sanitation Worker Hazards:** Extremely unsafe conditions for municipality workers handling manual touch sorting.



■ How Pasumai Madurai Works

Your Local AI Segregation Guide

- 1 Bilingual Interface (தமிழ் / English)**
Designed directly for Madurai residents with accessible terminology matching local habits.
- 2 Intuitive AI Search**
Enter any item in descriptive terms (e.g. "crushed bottle") to get correct disposal paths.
- 3 Actionable Step-by-Step Guide**
Get localized instructions (வழிமுறைகள்) detailing clean disposal along with ecological insights.



■ The Four-Bin Stream System

● Green Bin மட்கும் குப்பை

Organic Waste: Vegetable peels, organic leftovers, garden clippings, and banana leaves.

♻️ COMPOSTING STREAM

● Blue Bin மட்காத குப்பை

Dry Recyclables: Clean paper, hard cardboard, plastic bottles, metal cans, and dry covers.

♻️ PROCESSING CENTER

● Hazardous ஆபத்தானவை

Sanitary & Medical: Expired medicine strips, bandages, sanitary pads, and household chemicals.

⚠️ SAFE DISPOSAL

● E-Waste மின்னணுக் கழிவு

Circuitry & Electronics: Mobile chargers, broken appliances, remote cells, and dead batteries.

⚡ CERTIFIED DROP-OFFS

■ Intelligent User Experience



Localized Context

Features specific focus points designed for Madurai's geography, ensuring maximum safety for the iconic Vaigai River basin, local groundwater pools, and regional composting plants.



Smart Suggestions

Catches user typos and partial entries with instant, intuitive fallback paths. Interactive click-chips prompt helpful examples like "plastic cover" or "medicine pack."



Bilingual Integration

Seamlessly integrates deep local waste concepts in both English and colloquial Tamil, ensuring all household blocks can operate the assistant effectively.

■ Sustainability & Expected Impact



For the Environment

Converts organic food wastes to nutrient-rich compost, feeding local farms while recovering metals from electronic scrap.



For the Economy

Drastically slashes municipal carbon footprint, limits landfill overflow, and streamlines lucrative recycling lines.



For Society

Directly protects, dignifies, and safeguards the physical health of Madurai's crucial sanitation workforces.



■ Image Sources

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